

10.68 Find the Thevenin equivalent at terminals $a-b$ in the circuit of Fig. 10.111.

ML

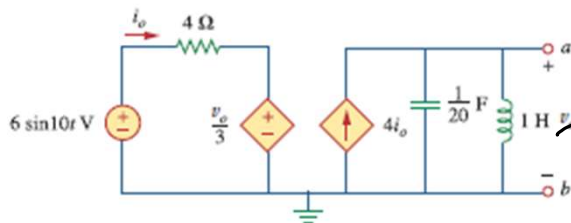


Figure 10.111
For Prob. 10.68.

$$V_{th} = V_{oc}$$

with dependent source 4

open terminal Z_{th}

①

Independent + sources

②

turn off
Add dummy
current / voltage
source in
OT.

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ML

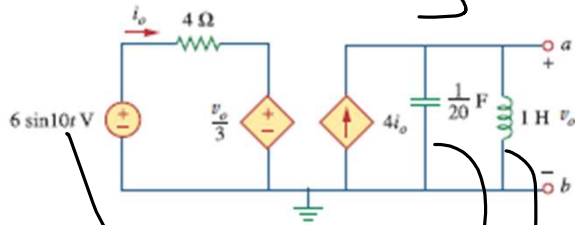


Figure 10.111
For Prob. 10.68.

$$\frac{V_{th}}{1H}$$

$$\rightarrow (w) j = 16 \times 1 \times j = j16$$

$$\frac{1}{20} \rightarrow -j \times \frac{1}{20} \rightarrow -j^2$$

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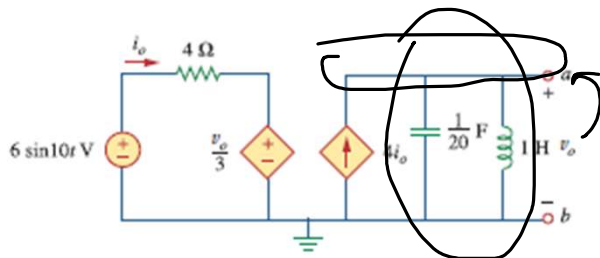


Figure 10.111

For Prob. 10.68.

$$V_{th} = j10 \parallel (-2j)$$

$$= -j2.5$$

$$V_o = 4i_o \times (-j2.5)$$

$$= -10 \times j \times i_o$$

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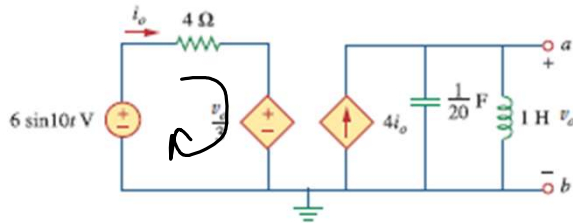


Figure 10.111
For Prob. 10.68.

$$-6 \angle 0^\circ + 4i_o + \frac{V_o}{3}$$

$$\Rightarrow 4i_o = \frac{V_o + 18 \angle 0^\circ}{12}$$

$$V_o = -10j \times \frac{V_o + 18 \angle 0^\circ}{12}$$

$$= -10j \times \frac{V_o + 18 \angle 0^\circ}{12}$$

$$\Rightarrow V_0 = \frac{-10j \times V_0 + 180 \angle -90^\circ}{12}$$

$$\Rightarrow 12V_0 = -10j V_0 + 180 \angle -90^\circ$$

$$\therefore V_0 = V_{th} = \frac{180 \angle -90^\circ}{12 + j10} = 11.52 \angle -130^\circ$$

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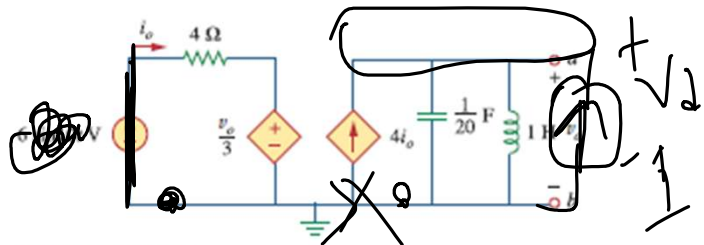


Figure 10.111
For Prob. 10.68.

Applying KCL at

$$4i_o + 12^{\circ} = \frac{v_d}{-j2.5}$$

$$\therefore i_o = \frac{v_d/3}{4}$$

$$= \frac{v_d}{12}$$

$$4 \times \frac{V_0}{12} + 1 \angle 0^\circ = \frac{V_0}{-j2.5}$$

$$V_0 = V_d$$

$$V_0 = \frac{1 \angle 0^\circ}{\left(\frac{1}{3} + \frac{1}{j2.5} \right)}$$

$$V_o = V_d = 1.92 \angle 50^\circ$$

$$Z_{tr} = \frac{V_d}{I_d} = \frac{1.92 \angle 50^\circ}{1 \angle 0^\circ} = 1.92 \angle 50^\circ$$

Practice Problem 10.6

Calculate v_o in the circuit of Fig. 10.15 using the superposition theorem.

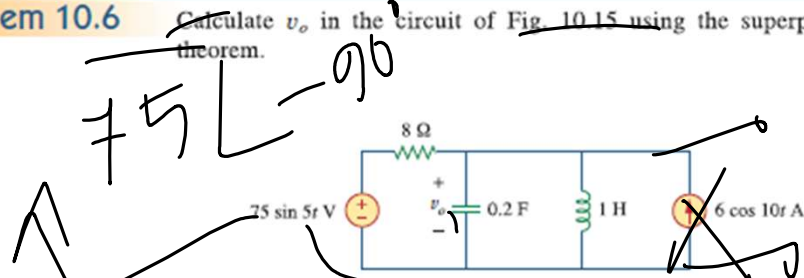


Figure 10.15
For Practice Prob. 10.6.

Use the following trigonometric identities:

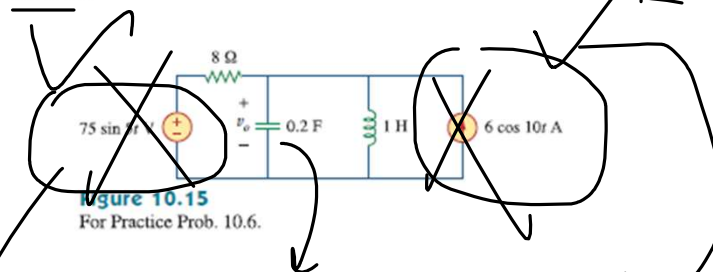
$$\begin{cases} \sin(\omega t \pm 180^\circ) = -\sin \omega t \\ \cos(\omega t \pm 180^\circ) = -\cos \omega t \\ \sin(\omega t \pm 90^\circ) = \pm \cos \omega t \\ \cos(\omega t \pm 90^\circ) = \mp \sin \omega t \end{cases}$$

Keeping $75 \angle 0^\circ$ ON
 V_{o1}
 $0.2 F$

$$\begin{aligned} & \rightarrow -j \frac{1}{0.2 \times 5} \\ & = -j \frac{1}{1} = -j \Omega \\ & 1 H \rightarrow j(5 \times 1) = j5 \Omega \end{aligned}$$

Practice Problem 10.6

Calculate v_o in the circuit of Fig. 10.15 using the superposition theorem.



$$V_o = V_{o1} + V_{o2}$$

$$j5 \parallel (-j) = -j1.25$$

$$V_{01} = 75 \angle -90^\circ \times \frac{-j1.25}{-j1.25 + 8}$$

$$= 11.54 \angle -171.12^\circ$$

Practice Problem 10.6

Calculate v_o in the circuit of Fig. 10.15 using the superposition theorem.

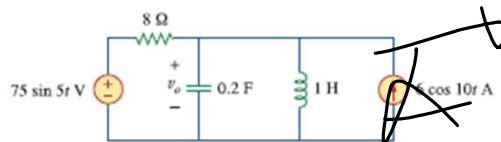


Figure 10.15
For Practice Prob. 10.6.

Keeping $6L^0$ ON

$$4 \parallel j5$$

$$= 4.25 \angle 58^\circ$$

Practice Problem 10.6

Calculate v_o in the circuit of Fig. 10.15 using the superposition theorem.

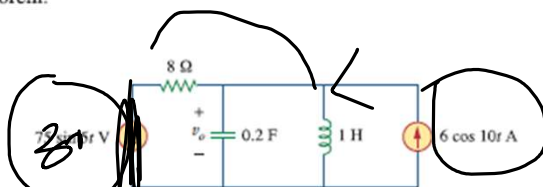


Figure 10.15
For Practice Prob. 10.6.

$$I^0 = \frac{4.25 \angle 58^\circ}{4.25 \angle 58^\circ - j} \times 6 \angle 0^\circ$$

$$= 1.23 \angle 8.9^\circ$$

$$V_{02} = i_{02} \times (-j)$$

$$= 1.23 \angle -81.15^\circ$$

$$V_0 = V_{01} + V_{02}$$

$$= 11.27 \angle 171.6^\circ$$

$$V_t = 11.54 \sin(5t - 81.11)$$

$$+ 12.5 \cos(10t - 81.11)$$

Norton

10.62 Using Thevenin's theorem, find v_o in the circuit of Fig. 10.105.

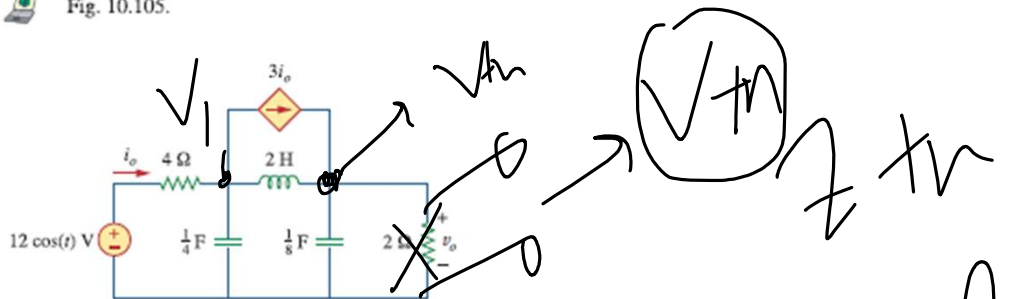


Figure 10.105
For Prob. 10.62.

Apply nodal analysis in
node $\rightarrow V_1$ & V_{th}
form KCL eqn

10.62 Using Thevenin's theorem, find v_o in the circuit of Fig. 10.105.

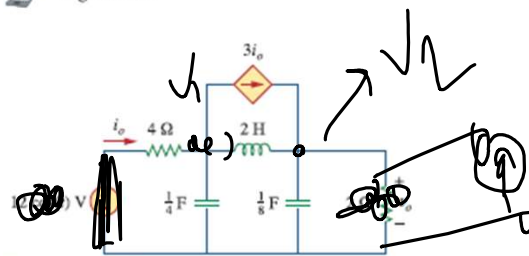


Figure 10.105
For Prob. 10.62.

$$1/4 F \rightarrow -j \times \frac{1}{1 \times 1/4}$$

$$\hookrightarrow -j4$$

$$-i_o + \frac{V_1}{-j4} + 3i_o + \frac{V_1 - V_2}{2j} = 0$$

$$i_o = \frac{V_1}{4}$$

$$\rightarrow -\frac{V_1}{4} + \frac{V_1}{-j4} + \frac{3}{4}V_1 + \frac{V_1 - V_2}{2j} = 0$$

10.62 Using Thevenin's theorem, find v_o in the circuit of Fig. 10.105.

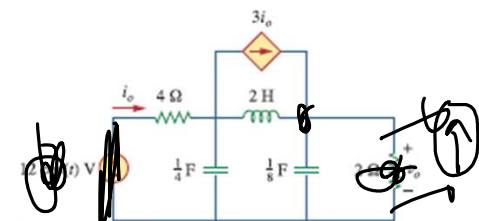


Figure 10.105
For Prob. 10.62.

$$12 \angle 0^\circ \text{ A}$$

$$\frac{3}{4} \frac{V_1}{2j} + \frac{V_1 - V_2}{2j}$$

$$- \frac{V_2}{j8} + \frac{12 \angle 0^\circ}{2} = 0$$

Solving for V_d

$$V_1 = -$$

$$V_2 = -$$

$$V_d = V_2$$

$$I_{th} =$$

$$\frac{V_d}{1 \angle 0^\circ}$$

- 10.66 At terminals $a-b$, obtain ~~the~~ Norton equivalent circuits for the network depicted in Fig. 10.109. Take $\omega = 10$ rad/s.

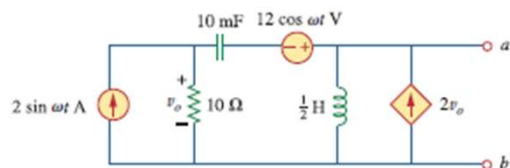


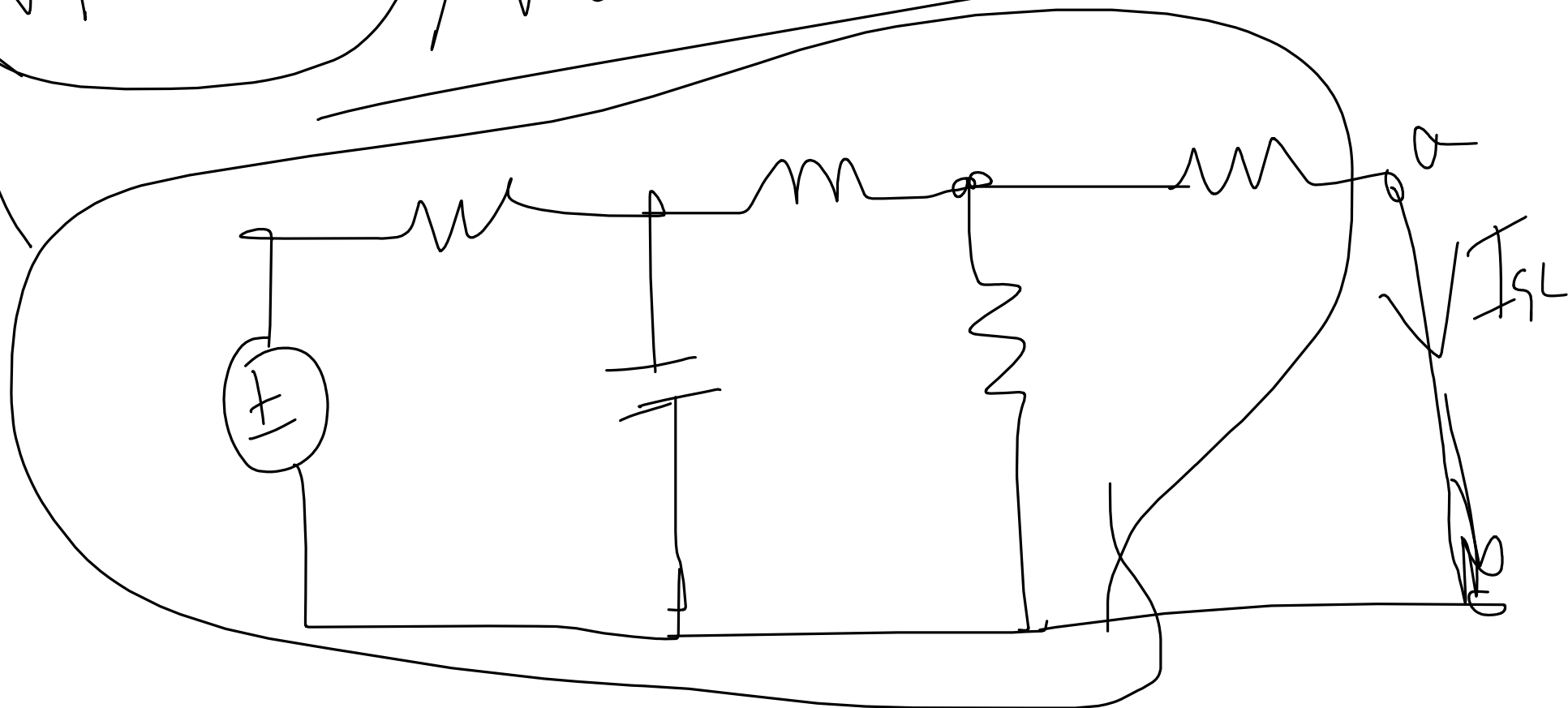
Figure 10.109
For Prob. 10.66.

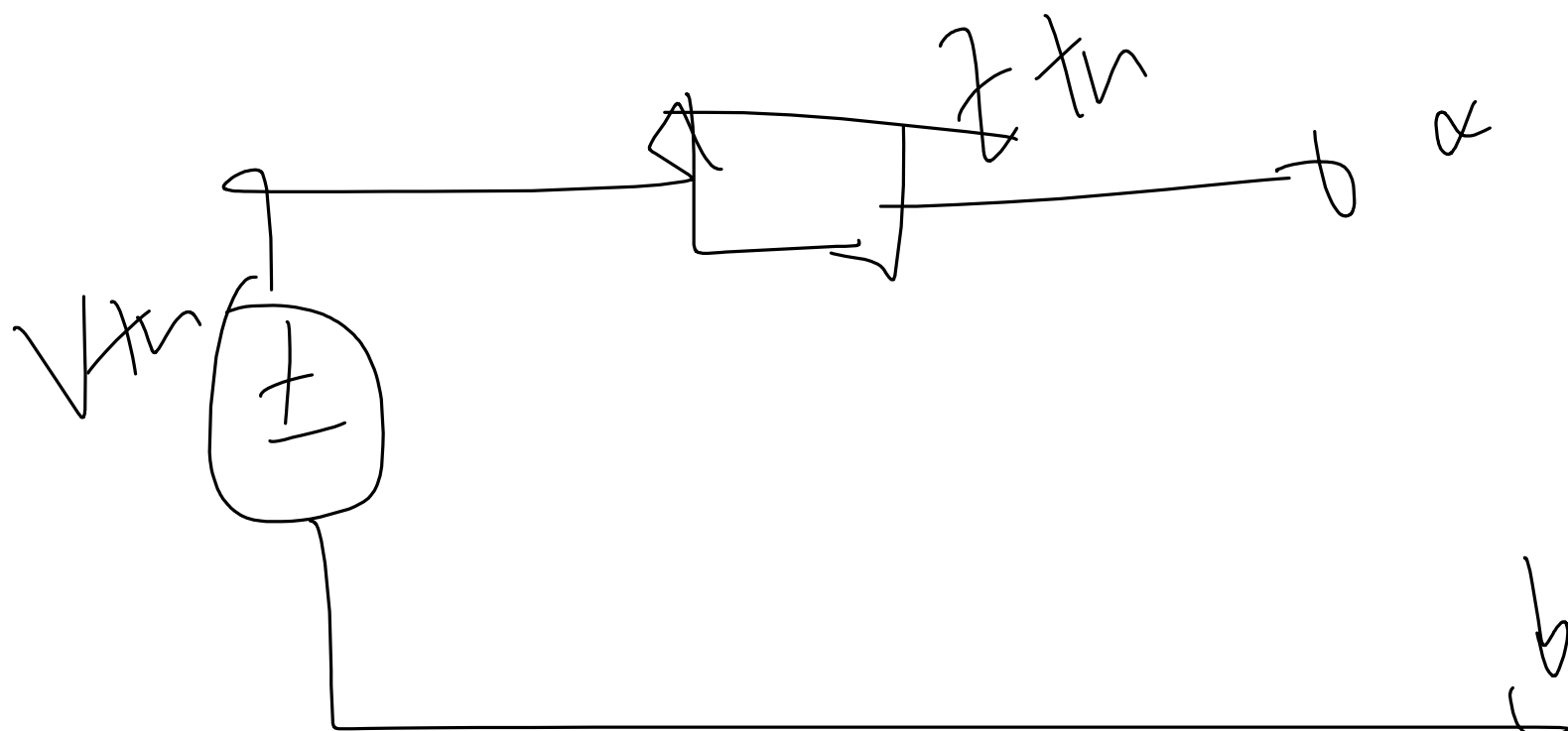
Solving KCL.

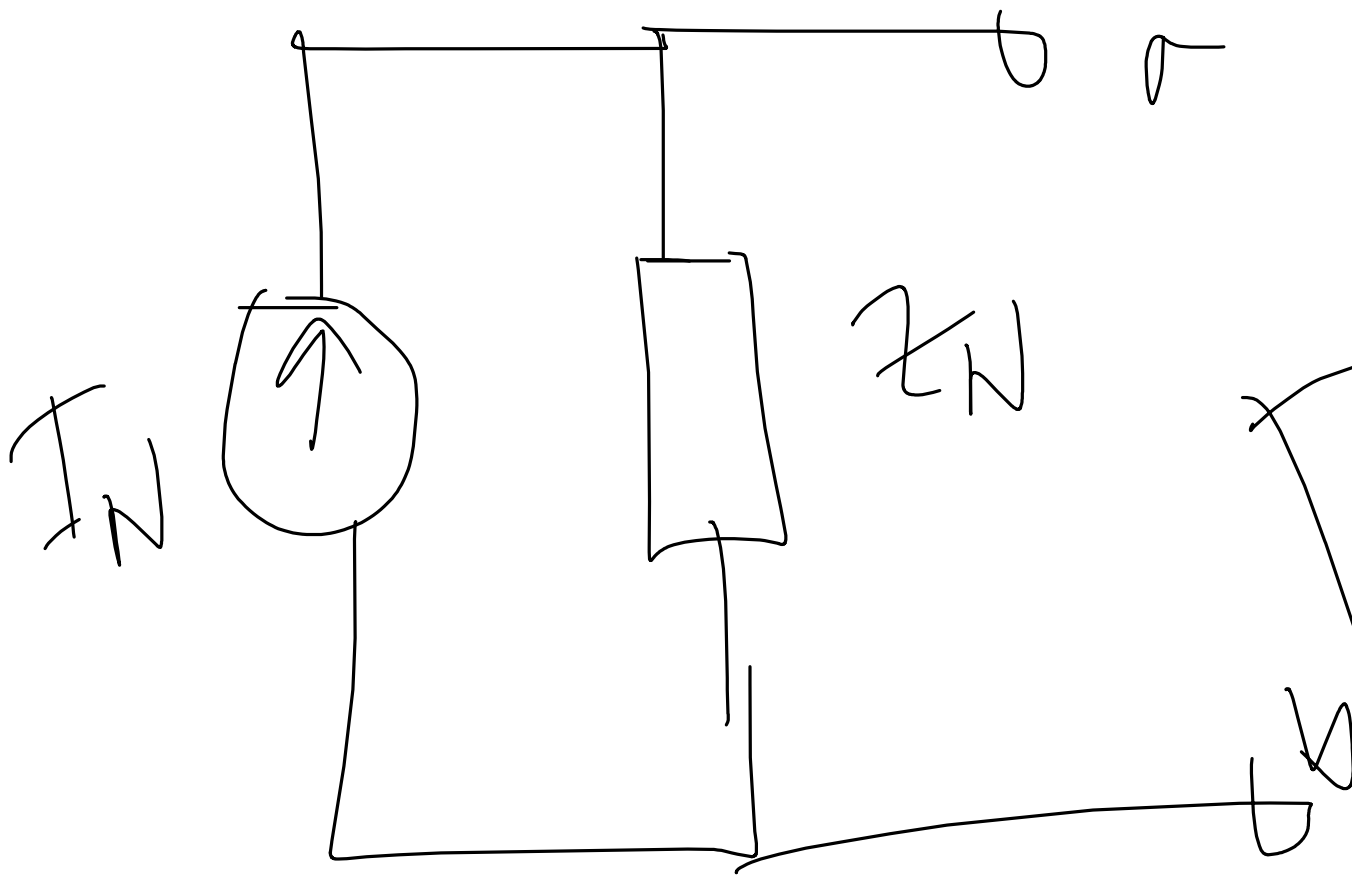
$V_1 = -$

$V_{th} = -$

V_{th}, Z_{th} Norton Theorem







$$I_N = I_{sc}$$

$$Z_N = Z_{th}$$

$$I_N = \frac{V_{th}}{Z_{th}}$$